SR UNIVERSITY

SchoolofComputerScienceandArtificialIntelligence

LabAssignment# 14

Name of Student :R.sai kundan

Enrollment No. : 2303A51413
Batch No. 21

TaskDescription1–(LandingPage Design)
• Createasimplelandingpageforaproductorservice.Includea header, hero section, and footer.
Requirements:
• UseHTMLandCSSto structure the page.
• UseAItosuggestlayoutalignment andtypography.
• Ensurethe pageisresponsiveacrossdevices.

Prompt:-
createa landing page forawebsite for aproduct orservice. It should havea header, amain sectionwithsomecontent,andafooter.UseappropriateHTMLtagsandstructuretocreate the page.

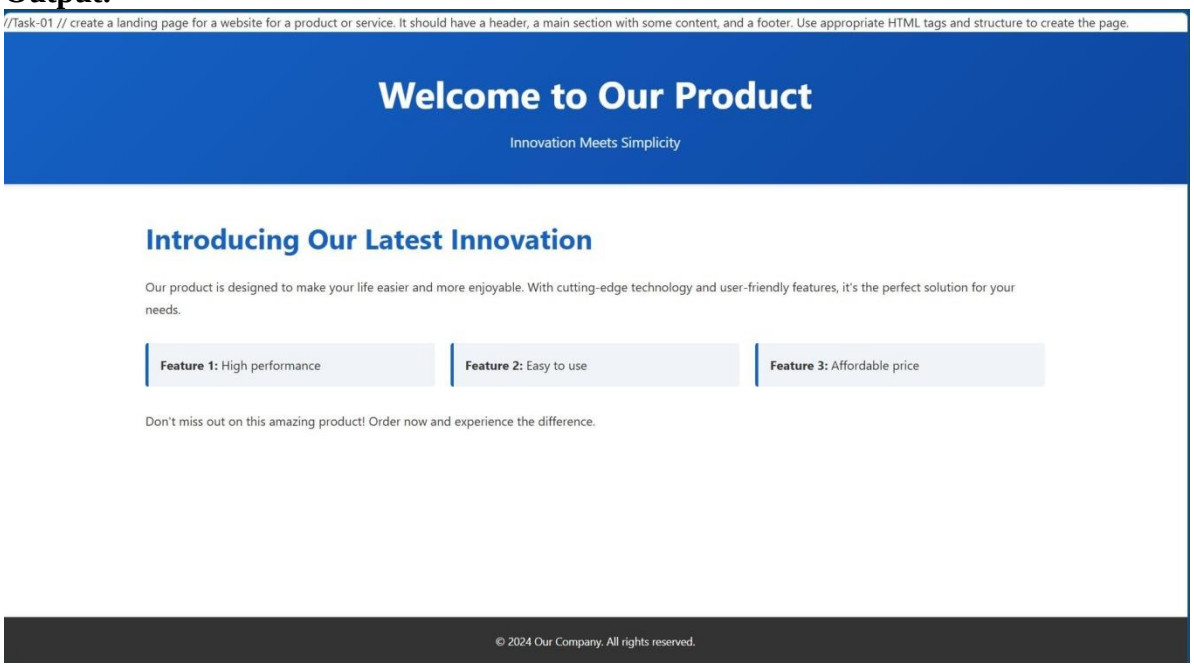
Code:-

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Product Landing Page</title>
    <style>
        * {
            margin: 0;
            padding: 0;
            box-sizing: border-box;
        }
        body {
            font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
            line-height: 1.6;
            color: #333;
        }
        header {
            background: linear-gradient(135deg, #1561c4 0%, #0d47a1 100%);
            color: white;
            padding: 40px 20px;
            text-align: center;
            box-shadow: 0 2px 5px rgba(0,0,0,0.1);
        }
        header h1 {
            font-size: clamp(24px, 5vw, 48px);
            margin-bottom: 10px;
        }
        header p {
            font-size: clamp(14px, 2vw, 18px);
            opacity: 0.9;
        }
        main {
            padding: 40px 20px;
            max-width: 1200px;
            margin: 0 auto;
            padding-bottom: 120px;
        }
        main h2 {
            font-size: clamp(20px, 4vw, 36px);
            margin-bottom: 20px;
            color: #1561c4;
```

```
}
main p {
    font-size: clamp(14px, 1.5vw, 16px);
    margin-bottom: 20px;
    line-height: 1.8;
}
ul {
    list-style: none;
    display: grid;
    grid-template-columns: repeat(auto-fit, minmax(250px, 1fr));
    gap: 20px;
    margin: 30px 0;
}
li {
    background: #f0f4f8;
    padding: 15px;
    border-left: 4px solid #1561c4;
    border-radius: 4px;
}
footer {
    background-color: #333;
    color: white;
    text-align: center;
    padding: 20px;
    position: fixed;
    width: 100%;
    bottom: 0;
    font-size: clamp(12px, 1vw, 14px);
}
@media (max-width: 768px) {
    main {
        padding: 30px 15px;
        padding-bottom: 100px;
    }
    ul {
        grid-template-columns: 1fr;
    }
}
```

```
    </style>
</head>
<body>
    <header>
        <h1>Welcome to Our Product</h1>
        <p>Innovation Meets Simplicity</p>
    </header>
    <main>
        <h2>Introducing Our Latest Innovation</h2>
        <p>Our product is designed to make your life easier and more enjoyable. With cutting-edge technology and user-friendly features,
        <ul>
            <li><strong>Feature 1:</strong> High performance</li>
            <li><strong>Feature 2:</strong> Easy to use</li>
            <li><strong>Feature 3:</strong> Affordable price</li>
        </ul>
        <p>Don't miss out on this amazing product! Order now and experience the difference.</p>
    </main>
    <footer>
        &copy; 2024 Our Company. All rights reserved.
    </footer>
</body>
</html>
```

Output:-



Justification:-
Showcasesaproductsectionwithclearheadingsandbulletpointstohighlightkeyfeatures and benefits for users. Focuses on readable layout and persuasive copy to improve user engagement and conversion.

TaskDescription2–(Theme Toggle Feature)
• Provideabuttontoswitchbetweenlightanddarkthemes..
Requirements:
• UseJavaScripttotoggleCSSclasses.
• UseAIassistance tomanagecolor contrast andaccessibility.
• Ensurethemepreference appliesinstantlyonclick.

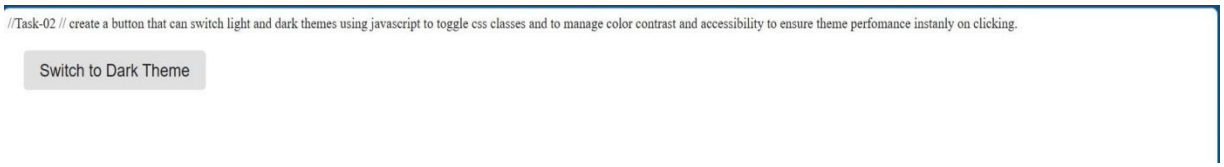
Prompt:-
createabuttonthatcanswitchlightanddarkthemesusingjavascripttotogglecssclassesand to manage color contrast and accessibility to ensure theme perfomance instanly on clicking.

Code:-

```html
<html>
    <head>
        <style>
            body.light-theme {
            }
            body.dark-theme {
                background-color: #222222;
                color: #ffffff;
            }
            .theme-toggle-btn {
                margin: 20px;
                padding: 10px 20px;
                font-size: 1.2em;
                cursor: pointer;
                border: none;
                border-radius: 5px;
                background: #e0e0e0;
                color: #222;
                transition: background 0.2s, color 0.2s;
            }
            body.dark-theme .theme-toggle-btn {
                background: #444;
                color: #fff;
            }
        </style>
    </head>
    <body class="light-theme">
        <button id="themeToggle" class="theme-toggle-btn" aria-pressed="false" aria-label="Switch to dark theme">
            Switch to Dark Theme
        </button>
        <script>
            const btn = document.getElementById('themeToggle');
            const body = document.body;
            function updateAccessibility(isDark) {
                btn.setAttribute('aria-pressed', isDark);
                btn.setAttribute('aria-label', isDark ? 'Switch to light theme' : 'Switch to dark theme');
                btn.textContent = isDark ? 'Switch to Light Theme' : 'Switch to Dark Theme';
            }
            btn.addEventListener('click', () => {
                const isDark = body.classList.toggle('dark-theme');
                body.classList.toggle('light-theme', !isDark);
                updateAccessibility(isDark);
            });
            // Set correct theme on load if preferring dark mode
            const prefersDark = window.matchMedia('(prefers-color-scheme: dark)').matches;
            if (prefersDark) {
                body.classList.remove('light-theme');
                body.classList.add('dark-theme');
                updateAccessibility(true);
            }
        </script>
    </body>
</html>
```

Output:-

InwhiteTheme



InBlack Theme

Justification:-

Implementsathemetogglebuttonthatinstantlyswitchesbetweenlightanddarkmodesusing class-based styling. Improves accessibility and user comfort by adjusting colors dynamically without reloading the page.

TaskDescription3–(UserFeedbackForm)

• Createafeedbackformwithfields:Username,Rating,Comments.

Requirements:

• UseJavaScriptto validaterequired fields.

• Displaysuccessorerrormessagesdynamically.

• UseAItoimproveformusabilityandlayout.

Prompt:-

createafeedbackformwithinfoofusername,rating,commentsusingjavascripttovalidate required field and usabilty and layout.

Code:-

```html
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Feedback Form</title>
</head>
<body>
  <div id="formContainer"></div>
  <script>
    // Create the form HTML structure dynamically
    const formContainer = document.getElementById('formContainer');
    formContainer.innerHTML = `
      <form id="feedbackForm" style="max-width:400px;margin:30px auto;padding:20px;border-radius:8px;box-shadow:0 2px 8px #ddd;background:#fafbfc;">
        <h2 style="text-align:center;">Feedback Form</h2>
        <div style="margin-bottom:15px;">
          <label for="username" style="display:block;margin-bottom:4px;">Username<span style="color:red;">*</span>:</label>
          <input type="text" id="username" name="username" required style="width:100%;padding:8px;border-radius:4px;border:1px solid #ccc;">
          <div id="usernameError" style="color:red;font-size:0.95em;display:none;margin-top:2px;"></div>
        </div>
        <div style="margin-bottom:15px;">
          <label for="rating" style="display:block;margin-bottom:4px;">Rating<span style="color:red;">*</span>:</label>
          <select id="rating" name="rating" required style="width:100%;padding:8px;border-radius:4px;border:1px solid #ccc;">
            <option value="">-- Select rating --</option>
            <option value="5">5 - Excellent</option>
            <option value="4">4 - Good</option>
            <option value="3">3 - Average</option>
            <option value="2">2 - Poor</option>
            <option value="1">1 - Terrible</option>
          </select>
          <div id="ratingError" style="color:red;font-size:0.95em;display:none;margin-top:2px;"></div>
        </div>
        <div style="margin-bottom:15px;">
          <label for="comments" style="display:block;margin-bottom:4px;">Comments:</label>
          <textarea id="comments" name="comments" rows="4" style="width:100%;padding:8px;border-radius:4px;border:1px solid #ccc;"></textarea>
        </div>
        <button type="submit" style="padding:10px 24px;border:none;border-radius:5px;background:#4CAF50;color:#fff;font-size:1.1em;cursor:pointer;">Submit</button>
        <div id="formFeedback" style="margin-top:15px;font-weight:bold;"></div>
      </form>
    `;
    // Form validation and usability
```

```
    const feedbackForm = document.getElementById('feedbackForm');
    const usernameInput = document.getElementById('username');
    const ratingInput = document.getElementById('rating');
    const commentsInput = document.getElementById('comments');
    const usernameError = document.getElementById('usernameError');
    const ratingError = document.getElementById('ratingError');
    const formFeedback = document.getElementById('formFeedback');
    function validateUsername() {
      if (usernameInput.value.trim() === "") {
        usernameError.textContent = "Username is required.";
        usernameError.style.display = "block";
        return false;
      } else {
        usernameError.textContent = "";
        usernameError.style.display = "none";
        return true;
      }
    }
    function validateRating() {
      if (!ratingInput.value) {
        ratingError.textContent = "Please select a rating.";
        ratingError.style.display = "block";
        return false;
      } else {
        ratingError.textContent = "";
        ratingError.style.display = "none";
        return true;
      }
    }
    // Accessibility: show errors live
    usernameInput.addEventListener('input', validateUsername);
    ratingInput.addEventListener('change', validateRating);
    feedbackForm.addEventListener('submit', function (e) {
      e.preventDefault();
      const isUsernameValid = validateUsername();
      const isRatingValid = validateRating();
      if (isUsernameValid && isRatingValid) {
        formFeedback.style.color = "green";
        formFeedback.textContent = "Thank you for your feedback!";
        feedbackForm.reset();
        usernameError.style.display = "none";
        ratingError.style.display = "none";
        setTimeout(() => {formFeedback.textContent = "";}, 3000);
      } else {
        formFeedback.style.color = "red";
        formFeedback.textContent = "Please correct the errors above and submit again.";
      }
    });
  </script>
</body>
</html>
```

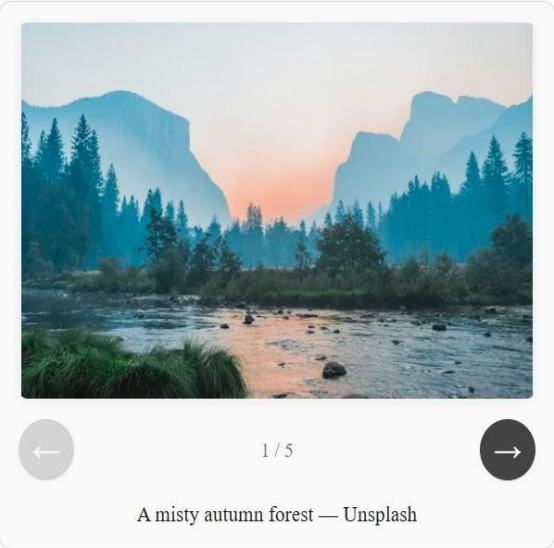
Output:-

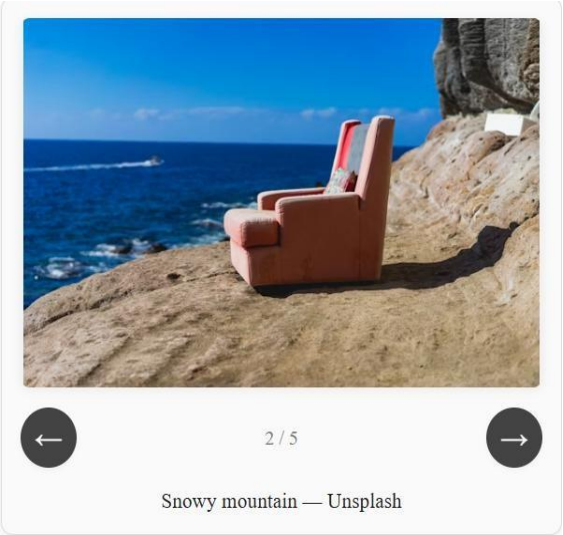
Justifications:-
Buildsafeedbackformcapturingusername,rating,andcommentswithJavaScriptvalidation for required fields. Enhances usability by preventing incomplete submissions and guiding users to correct input errors.

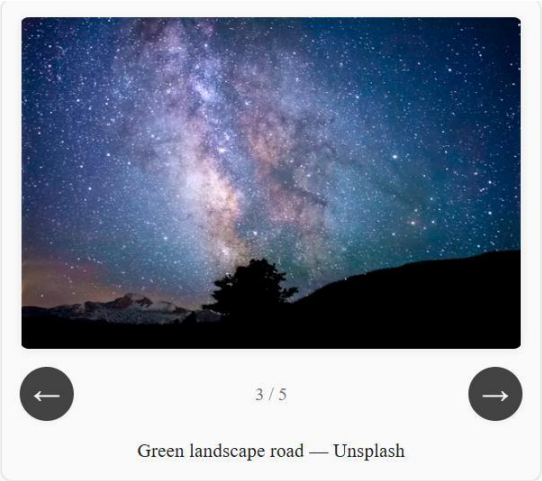
TaskDescription- 4(Image Gallery withJavaScript)
• Task:Displayasmallimagegalleryonawebpage.
Requirements:
• UseJavaScripttodynamicallyloadimages.
• Addnext/previousnavigationbuttons.
• UseAIto optimizelayout responsiveness.

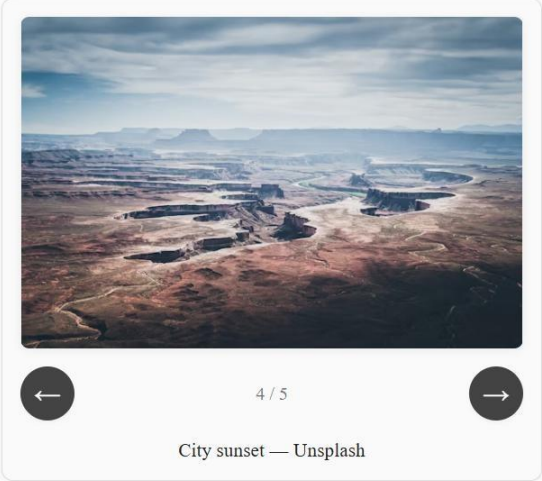
Prompt:-
createasmallimagegalleryonawebpageusinghtml,jstodynamicallyloadimagesandadd navigations on next/previous buttons and optimize layout responsiveness, HTML structure for the image gallery

Output:-









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Justification:-

Createsaresponsiveimagegallerythatdynamicallyloadsimagesandsupportsnext/previous navigation buttons.Provides a clean, interactive browsing experience optimized for different screen sizes

Code:

```html
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Responsive JavaScript Image Gallery</title>
  <style>
    :root {
      --gallery-max-width: 480px;
      --gallery-margin: 0 auto;
      --gallery-bg: #fafafa;
      --gallery-border: 1px solid #ddd;
      --gallery-radius: 10px;
      --btn-bg: #333;
      --btn-color: #fff;
      --btn-radius: 50%;
      --btn-size: 48px;
    }

    .gallery-container {
      max-width: var(--gallery-max-width);
      margin: var(--gallery-margin);
      background: var(--gallery-bg);
      border: var(--gallery-border);
      border-radius: var(--gallery-radius);
      padding: 16px;
      box-sizing: border-box;
      display: flex;
      flex-direction: column;
      align-items: center;
      box-shadow: 0 2px 10px 0 rgba(0,0,0,0.08);
    }
    .gallery-image-wrapper {
      position: relative;
      width: 100%;
      padding-top: 66%; /* Optimized for 3:2 ratio fo
      background: #f1f2f4;
      border-radius: 8px;
      overflow: hidden;
      margin-bottom: 16px;
      transition: box-shadow .18s;
      box-shadow: 0 2px 8px 0 rgba(0,0,0,.06);
      display: flex;
      align-items: center;
      justify-content: center;
    }
    .gallery-image-wrapper img {
      margin: 0 auto;
      position: absolute;
      left: 0; top: 0; width: 100%; height: 100%;
      object-fit: contain;
      transition: opacity .16s;
      user-select: none;
      pointer-events: none;
    }
```

```css
.gallery-buttons {
  display: flex;
  justify-content: space-between;
  align-items: center;
  margin-bottom: 8px;
}
.gallery-btn {
  background: var(--btn-bg);
  color: var(--btn-color);
  border: none;
  border-radius: var(--btn-radius);
  width: var(--btn-size);
  height: var(--btn-size);
  font-size: 1.5em;
  cursor: pointer;
  transition: background .18s, filter .18s;
  display: flex;
  align-items: center;
  justify-content: center;
  outline: none;
  opacity: 0.92;
  box-shadow: 0 2px 4px #acacac22;
  z-index: 2;
}
.gallery-btn:disabled {
  background: #bbb;
  cursor: not-allowed;
  opacity: 0.6;
}

.gallery-caption {
  font-size: 1.07em;
  text-align: center;
  margin-top: 0.5em;
  color: #222;
}
@media (max-width: 600px) {
  .gallery-container {
    padding: 8px;
    max-width: 98vw;
  }
  .gallery-image-wrapper {
    padding-top: 60%;
    border-radius: 6px;
  }
  .gallery-btn {
    width: 40px;
    height: 40px;
    font-size: 1.1em;
  }
  .gallery-caption {
    font-size: 0.97em;
  }
}
@media (max-width: 400px) {
  .gallery-btn {
    width: 36px;
    height: 36px;
    font-size: 0.95em;
  }
  .gallery-image-wrapper {
    padding-top: 56%;
  }
```

```html
 </style>
</head>
<body>
 <div class="gallery-container">
   <div class="gallery-image-wrapper" id="imgWrapper">
     <!-- The image will be inserted here by JS -->
   </div>
   <div class="gallery-buttons">
     <button class="gallery-btn" id="prevBtn" aria-label="Previous image">&larr;</button>
     <span id="galleryCounter" style="min-width: 50px;text-align: center;font-size: 1em; color: #777"></span>
     <button class="gallery-btn" id="nextBtn" aria-label="Next image">&rarr;</button>
   </div>
   <div class="gallery-caption" id="imgCaption"></div>
 </div>
 <script>
   // AI optimized: Use small, modern placeholder pics & alt text for a11y
   const images = [
     {
       src: "https://images.unsplash.com/photo-1506744038136-46273834b3fb?auto=format&fit=crop&w=600&q=80",
       alt: "A forest in autumn morning mist",
       caption: "A misty autumn forest — Unsplash"
     },
     {
       src: "https://images.unsplash.com/photo-1519125323398-675f0ddb6308?auto=format&fit=crop&w=600&q=80",
       alt: "Snowy mountain peak with clouds",
       caption: "Snowy mountain — Unsplash"
     },
     {
       src: "https://images.unsplash.com/photo-1465101046530-73398c7f28ca?auto=format&fit=crop&w=600&q=80",
       alt: "Winding road in green landscape",
       caption: "Green landscape road — Unsplash"
     },
     {
       src: "https://images.unsplash.com/photo-1519985176271-adb1088fa94c?auto=format&fit=crop&w=600&q=80",
       alt: "City skyline at sunset",
       caption: "City sunset — Unsplash"
     },
     {
       src: "https://images.unsplash.com/photo-1502082553048-f009c37129b9?auto=format&fit=crop&w=600&q=80",
       alt: "Closeup of sunflower",
       caption: "Sunflower closeup — Unsplash"
     }
   ];

   let currentIdx = 0;

   const imgWrapper = document.getElementById('imgWrapper');
   const prevBtn = document.getElementById('prevBtn');
   const nextBtn = document.getElementById('nextBtn');
   const imgCaption = document.getElementById('imgCaption');
   const galleryCounter = document.getElementById('galleryCounter');
```

```html
function updateGallery(idx) {
  // Remove previous img
  imgWrapper.innerHTML = "";
  // Responsive: use srcset for layout and bandwidth optimization
  const img = document.createElement('img');
  img.src = images[idx].src;
  img.alt = images[idx].alt;
  img.decoding = "async";
  img.loading = "lazy";
  img.srcset = images[idx].src.replace("w=600", "w=400") + " 400w, " +
               images[idx].src + " 600w";
  img.sizes = "(max-width: 480px) 98vw, 480px";
  imgWrapper.appendChild(img);

  imgCaption.textContent = images[idx].caption;
  galleryCounter.textContent = (idx+1) + " / " + images.length;

  prevBtn.disabled = idx === 0;
  nextBtn.disabled = idx === images.length - 1;
}
// Handle navigation
prevBtn.addEventListener('click', () => {
  if (currentIdx > 0) {
    currentIdx--;
    updateGallery(currentIdx);
  }
});
nextBtn.addEventListener('click', () => {
  if (currentIdx < images.length - 1) {
    currentIdx++;
    updateGallery(currentIdx);
  }
});

// Keyboard accessibility
document.addEventListener('keydown', e => {
  if (e.key === "ArrowLeft" && currentIdx > 0) {
    currentIdx--;
    updateGallery(currentIdx);
  }
  if (e.key === "ArrowRight" && currentIdx < images.length - 1) {
    currentIdx++;
    updateGallery(currentIdx);
  }
});

// Initial load
updateGallery(currentIdx);
</script>
</body>
</html>
```

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TaskDescription-5:(Notification Banner)
• Displayanotificationbanneratthetopofthepage.
Requirements:
• UseJavaScripttoshowandhidethebanner.
• StylethebannerusingCSS.
• UseAIto ensuresmoothtransitionsandcleanUIbehavior.

Prompt:-
createawebpagetodispalythenotificationbanneratthetopofthepageusinghtml,jsto showthehiddenthebannerandstylethebannerusingcsstoensuresmoothtransitionand clean UI behaviour.

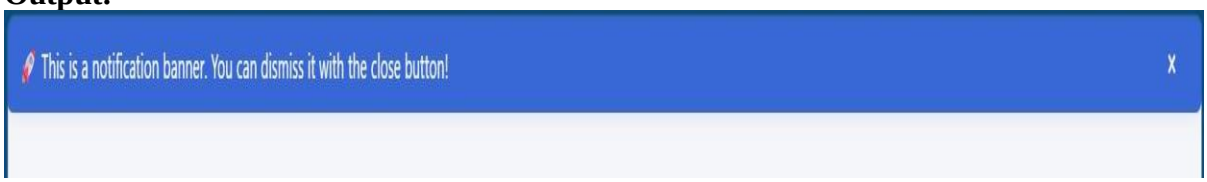
Code:

```html
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Notification Banner Example</title>
  <style>
    :root {
      --banner-bg: #3867d6;
      --banner-color: #fff;
      --banner-height: 48px;
      --banner-font: 1.05em;
      --banner-radius: 0 0 7px 7px;
      --banner-shadow: 0 2px 12px rgba(44,62,80,0.08);
      --banner-transition: transform 0.35s cubic-bezier(.8,.22,.23,.89), opacity .25s;
      --close-btn-size: 28px;
      --close-btn-bg: transparent;
      --close-btn-hover: #ffff5;
      --close-btn-color: #fff;
    }
    body {
      margin: 0;
      font-family: system-ui,sans-serif;
      background: #f5f6fa;
      min-height: 400px;
    }
    .notif-banner {
      box-sizing: border-box;
      width: 100%;
      background: var(--banner-bg);
      color: var(--banner-color);
      min-height: var(--banner-height);
      display: flex;
      align-items: center;
      justify-content: space-between;
      position: fixed;
      left: 0;
      top: 0;
      z-index: 1000;
      border-radius: var(--banner-radius);
      box-shadow: var(--banner-shadow);
      padding: 0 24px 0 16px;
      font-size: var(--banner-font);
      letter-spacing: 0.01em;
      transform: translateY(-110%);
      opacity: 0;
      pointer-events: none;
      transition: var(--banner-transition);
    }
    .notif-banner.show {
      transform: translateY(0);
      opacity: 1;
      pointer-events: auto;
    }
```

```html
    .notif-banner__text {
      flex: 1;
      padding: 6px 0;
      text-align: left;
    }
    .notif-banner__close {
      appearance: none;
      border: none;
      background: var(--close-btn-bg);
      color: var(--close-btn-color);
      border-radius: 50%;
      width: var(--close-btn-size);
      height: var(--close-btn-size);
      font-size: 1.22em;
      cursor: pointer;
      display: flex;
      align-items: center;
      justify-content: center;
      transition: background 0.18s;
      outline: none;
      margin-left: 14px;
    }
    .notif-banner__close:focus,
    .notif-banner__close:hover {
      background: var(--close-btn-hover);
    }

    /* Display 'Show' button only when bann
    #showBannerBtn {
      position: fixed;
      top: 14px;
      right: 16px;
      z-index: 900;
      padding: 7px 16px;
      background: #fff;
      border: 1px solid #3867d6;
      color: #3867d6;
      border-radius: 18px;
      font-size: 1em;
      font-weight: 500;
      cursor: pointer;
      box-shadow: 0 0 8px #3867d610;
      opacity: 0;
      pointer-events: none;
      transition: opacity 0.22s;
    }
    #showBannerBtn.active {
      opacity: 1;
      pointer-events: auto;
    }
    @media (max-width: 480px) {
      .notif-banner {
        font-size: .95em;
        padding-right: 8px;
      }
      #showBannerBtn {
        font-size: .97em;
        right: 8px;
      }
    }
  </style>
</head>
<body>
```

```html
<body>
  <div class="notif-banner" id="notifBanner" aria-live="polite" role="status">
    <span class="notif-banner__text">🚀 This is a notification banner. You can dismiss it with the close button!</span>
    <button class="notif-banner__close" id="closeBannerBtn" aria-label="Dismiss notification">&times;</button>
  </div>
  <button id="showBannerBtn">Show Notification</button>

  <script>
    // Elements
    const banner = document.getElementById('notifBanner');
    const closeBtn = document.getElementById('closeBannerBtn');
    const showBtn = document.getElementById('showBannerBtn');

    // Show the banner on page load (AI/UX: small delay to catch the eye)
    setTimeout(() => {
      banner.classList.add('show');
      showBtn.classList.remove('active');
    }, 250);

    // Hide the banner
    closeBtn.addEventListener('click', () => {
      banner.classList.remove('show');
      // Show the 'show' button for accessibility
      setTimeout(() => showBtn.classList.add('active'), 400);
    });

    // Show the banner again when user clicks 'Show Notification'
    showBtn.addEventListener('click', () => {
      banner.classList.add('show');
      showBtn.classList.remove('active');
    });

    // Optional: Hide banner on Escape key for a11y
    document.addEventListener('keydown', e => {
      if (e.key === "Escape" && banner.classList.contains('show')) {
        banner.classList.remove('show');
        setTimeout(() => showBtn.classList.add('active'), 400);
      }
    });
  </script>
</body>
</html>
```

Output:-



Justification:-

Displays a notification banner at the top of the page that can be shown or hidden using JavaScript.Usessmoothtransitionsandstyledlayouttodeliverimportantmessageswithout disrupting the overall UI.